iLearn Practice Exam

10BR · Chapter / Topic 4

Cambridge IGCSE Computer Science 0478

Paper 1 Computer Systems

Chapter 4: Software

Name: _____

Class: _____

Date: _____

Time allowed: 45 minutes

TOTAL MARKS: 39

Marks obtained: ..... / 39

INSTRUCTIONS

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

INFORMATION

· The total mark for this paper is 39.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 5 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, your teacher will mark your work against the official mark scheme.
· Only 39 marks of past-paper questions are available for this chapter, so the recommended time is 45 minutes (1 min 1

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1 A student uses both system software and application software on their computer.

(a) Give **one** example of system software.

..... [1]

(b) Give **two** examples of application software.

1 .....

2 .....
[2]

(c) Describe the difference between system software and application software.

.....

.....

.....

.....

..... [2]

2 A programmer writes a program to store and sort all the data for the books in her home library.

(a) The programmer uses a high-level language to write the program.

One reason for writing the program using a high-level language is that it is easier to read and write.

Give **two** other reasons why the programmer chose to use a high-level language.

1 .....

.....

2 .....

.....
[2]

(b) The programmer uses an integrated development environment (IDE) to write the program.

One function of the IDE is to provide a translator.

(i) Complete the statements about translators.

Use the terms from the list. Some of the terms will **not** be used. You should only use a term once.

all together	all	assembler	binary	error
executable file	executing	interrupt	line by line	
partial code	some	translating	whole code	

A compiler translates the ..... at once before

..... it. A compiler produces an error report that displays

..... errors.

An interpreter translates and executes the code ..... .

An interpreter stops execution when an ..... is found

and continues once it is corrected.

[5]

(ii) Give **two** other functions commonly found in an IDE.

State the role of each function.

Function 1 .....

Role .....

.....

Function 2 .....

Role .....

.....

[4]

3 An optical mouse is connected to a computer using a universal serial bus (USB) cable.

(a) One function performed by the operating system is managing peripherals. A role of this function is to install the drivers needed for the optical mouse.

Complete the table with the missing functions and roles of an operating system.

function	role
providing an interface	....
....	It makes sure that two processes do **not** access the same memory location at the same time.
managing interrupts	....
....	It allows the user to set a username and password.

[4]

4 An integrated development environment (IDE) has two types of translator: a compiler and an interpreter.

(a) Give the purpose of the compiler and the interpreter.

.....

..... [1]

(b) Identify **one** other type of translator.

..... [1]

(c) The table contains statements about compilers and interpreters.

letter	statement
A	An interpreter translates and executes the code one variable at a time.
B	A compiler displays an error report when it is unable to run the code due to syntax errors.
C	An interpreter stops translating as soon as it finds an error in the code.
D	A compiler translates the whole code at once and then it is executed.
E	A compiler produces a non-executable file.

Two of the statements in the table are **not** correct.

Give the letter of each incorrect statement.

Suggest a corrected statement for each.

letter of the incorrect statement 1 .....

corrected statement .....

.....

.....

letter of the incorrect statement 2 .....

corrected statement .....

.....

.....

[4]

(d) A translator is one function of an IDE.

Identify **two** other functions of an IDE.

State the purpose of each function.

function 1 .....

purpose .....

.....

.....

function 2 .....

purpose .....

.....

.....

[4]

5 A programmer uses a low-level language to write a program for a vending machine.

(a) State what is meant by a low-level language.

.....

..... [1]

(b) Identify **two** examples of a low-level language.

1 .....

2 .....
[2]

(c) Give an example of a translator that can be used for a low-level language.

..... [1]

(d) State **two** reasons why the programmer chose a low-level language instead of a high-level language to write the program.

1 .....

.....

2 .....

.....
[2]

(e) The programmer uses an integrated development environment (IDE) to write the program.

Identify **three** terms from the given list that are functions of an IDE.

auto-correction	binary converter
internet protocol (IP) address	media access control (MAC) address
machine independence	prettyprint
protocol identifier	run-time environment

1 .....

2 .....

3 .....
[3]

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END OF QUESTION PAPER